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1. (a)
$$P_R(r) = \begin{cases} 0, & r < 0 \\ \binom{n}{r} p^r (1-p)^{n-r}, & 0 \leq r \leq n \\ 0, & r > n \end{cases}, \quad E[R] = np$$

$$E[R^2] = \sum_{r=0}^n r^2 \cdot \frac{n!}{r!(n-r)!} p^r (1-p)^{n-r} = \sum_{r=1}^n \frac{nr \cdot (n-1)!}{(r-1)!(n-r)!} p^r (1-p)^{n-r}$$

$$= np \sum_{r=0}^{n-1} \frac{(r+1)(n-1)!}{r!(n-1-r)!} p^r (1-p)^{n-1-r}$$

$$= np \cdot ((n-1)p + 1) = n(n-1)p^2 + np$$

$$\text{var}(R) = (n^2 - n - n^2)p^2 + np = np(1-p)$$

(b) Denote the first item to be loaded ends up being the only one on ~~red truck~~ ^{its truck is A}

~~A = A~~
$$P(A) = P(\text{2nd} \sim \text{50th} \text{ packages are loaded in green truck} \mid \text{1st package is load in red truck})$$

$$+ P(\text{2nd} \sim \text{50th} \text{ packages are loaded in red truck} \mid \text{1st package is loaded in green truck})$$

$$\times P(\text{1st package is loaded in red truck})$$

$$\times P(\text{1st package is loaded in green truck})$$

$$= (1-p)^{n-1} p + p^{n-1} (1-p)$$

(c) ~~Denote the event that red truck~~

~~R = 1~~

$$R + G = n. \text{ When } n \geq 3, P(R=1 \cap G \geq 1) = 0$$

$$P(\min(R, G) = 1) = P_R(1) + P_G(1) = n[p(1-p)^{n-1} + (1-p)p^{n-1}]$$

$$\text{When } n=2, P(R=G=1) = 2p(1-p) = 2p(1-p)$$

$$\text{When } n=1, P(R=1 \text{ or } G=1) = 1$$

(d) $E[R-G] = E[R+G] - 2E[G] = n - 2n(1-p) = n(2p-1)$

$$\text{var}(R-G) = \text{var}(R+R-n) = \text{var}(2R-n) = 4\text{var}(R) = 4np(1-p)$$

(e) Denote the first two packages to be loaded go onto the red truck is A
the event that

~~R = 2~~
 Denote $R' = R-2$, $P_{R'|A}(r) = \begin{cases} 0, & r < 0 \\ \binom{n-2}{r} p^r (1-p)^{n-2-r}, & 0 \leq r \leq n-2 \\ 0, & r > n-2 \end{cases} \Rightarrow P_{R|A}(r) = \begin{cases} 0, & r < 2 \\ \binom{n-2}{r-2} p^{r-2} (1-p)^{n-r}, & 2 \leq r \leq n \\ 0, & r > n \end{cases}$

$$\cancel{E[R] = E[R'+2] = 2 + E[R'] = 2 + (n-2)p}$$

$$E[R|A] = E[R'+2] = 2 + (n-2)p$$

$$\text{var}(R|A) = \text{var}(R'+2) = \text{var}(R') = (n-2)p(1-p)$$

2. Denote N_A is the number of the messages sent by A
 N_B by B

$$\begin{aligned} (a) P(9 \text{ messages are received}) &= \sum_{k=0}^9 P_A(k, t) P_B(9-k, t) \\ &= \sum_{k=0}^9 e^{-\lambda_A t} \frac{(\lambda_A t)^k}{k!} \cdot e^{-\lambda_B t} \frac{(\lambda_B t)^{9-k}}{(9-k)!} \\ &= \sum_{k=0}^9 e^{-(\lambda_A + \lambda_B)t} \frac{(\lambda_A t)^k (\lambda_B t)^{9-k}}{k! (9-k)!} = e^{-(\lambda_A + \lambda_B)t} \frac{[(\lambda_A + \lambda_B)t]^9}{9!} \end{aligned}$$

(b) Denote X = the number of messages are sent in duration t

W_i = the number of word in i th letter

$$W = W_1 + W_2 + \dots + W_X$$

$$E[W] = E[E[W|X]] = E[X E[W]] = E[X] E[W]$$

$$= (\lambda_1 + \lambda_2)t \times \frac{11}{6}$$

(c) ~~Denote~~ Denote T = the time that the receiver has received 8 words messages from A.

T has Erlang distribution with $k=8$, $\lambda = \frac{\lambda_A}{6}$

$$\cancel{f_T(t) = 1 - P(T > t) =}$$

$$\cancel{P(T \leq t) =}$$

$$\cancel{P_T(t)}$$

$$f_T(t) = \frac{(\frac{\lambda_A}{6})^8 t^7 e^{-\frac{\lambda_A}{6}t}}{7!}$$

$$(d) P = \binom{12}{8} \left(\frac{\lambda_A}{\lambda_A + \lambda_B} \right)^8 \left(\frac{\lambda_B}{\lambda_A + \lambda_B} \right)^4$$



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3. Denote X_1, X_2, X_3 are the service times of 3 customers
 $\uparrow$
 (me)

$$Y = X_1 - X_2$$

$$f_Y(y) = \int_0^{\infty} f_{X_1}(z+y) f_{X_2}(z) dz, (y > 0)$$

$$= \int_0^{\infty} \lambda_1 e^{-\lambda_1(z+y)} \lambda_2 e^{-\lambda_2 z} dz$$

$$= e^{-\lambda_1 y} \cdot \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

$$f_Y(y) = \int_0^{\infty} f_{X_1}(z) f_{X_2}(z-y) dz$$

$$= \int_0^{\infty} \lambda_1 e^{-\lambda_1 z} \lambda_2 e^{-\lambda_2(z-y)} dz = e^{\lambda_2 y} \cdot \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} (y < 0)$$

$$\text{if } Y > 0, P(X_3 < Y) = \int_0^Y f_{X_3}(z) dz = 1 - e^{-\lambda_3 Y}$$

$$\int_0^{\infty} P(X_3 < Y | Y=y > 0) f_Y(y) dy = \int_0^{\infty} (1 - e^{-\lambda_3 y}) \cdot e^{-\lambda_1 y} \cdot \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} dy$$

$$= \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_1 + \lambda_3} \right)$$

$$\text{obviously, } \int_{-\infty}^0 P(X_3 > Y | Y=y < 0) f_Y(y) dy = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} \left(\frac{1}{-\lambda_2} - \frac{1}{\lambda_1 + \lambda_3} \right)$$

$$\Rightarrow P(\text{won't be the last leave}) = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2} \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{2}{\lambda_1 + \lambda_3} \right)$$

$$= 1 - \frac{2\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2}$$

$$\Leftrightarrow P(\text{be the last leave}) = \frac{2\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2}$$

$$4. \frac{P(2, \frac{1}{3}) P(0, \frac{2}{3})}{P(2, 1)} = \frac{e^{(-\frac{\lambda}{3})} \cdot \frac{(\frac{\lambda}{3})^2}{2!} \cdot e^{-\frac{2}{3}\lambda}}{e^{-\lambda} \cdot \frac{\lambda^2}{2!}} = \frac{1}{9}$$